Unit M3 Mechanics 3

GCE AS and GCE Further Mathematics and GCE AS and GCE Further Mathematics (Additional) A2 optional unit

M3.1 Unit description

Further kinematics; elastic strings and springs; further dynamics; motion in a circle; statics of rigid bodies.

M3.2 Assessment information

Prerequisites

A knowledge of the specifications for M1 and M2 and their prerequisites and associated formulae, together with a knowledge of differentiation, integration and differential equations, as specified in C1, C2, C3 and C4, is assumed and may be tested.

Examination

The examination will consist of one 1½ hour paper. The paper will contain about seven questions with varying mark allocations per question which will be stated on the paper. All questions may be attempted.

Calculators

Students are expected to have available a calculator with at least the following keys: $+, -, \times, \div, \pi, x^2, \sqrt{x}, \frac{1}{x}, x^y, \ln x, e^x$, sine, cosine and tangent and their inverses in degrees and decimals of a degree, and in radians; memory. Calculators with a facility for symbolic algebra, differentiation and/or integration are not permitted.

Formulae

Students are expected to know any other formulae which might be required by the specification and which are not included in the booklet, *Mathematical Formulae including Statistical Formulae and Tables*, which will be provided for use with the paper. Questions will be set in SI units and other units in common usage.

Students will be expected to know and be able to recall and use the following formulae:

The tension in an elastic string = $\dfrac{\lambda x}{l}$

The energy stored in an elastic string = $\dfrac{\lambda x^2}{2l}$

For SHM:

$\ddot{x} = -\omega^2 x$,

$x = a \cos \omega t$ or $x = a \sin \omega t$,

$v^2 = \omega^2(a^2 - x^2)$,

$T = \dfrac{2\pi}{\omega}$

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1 Further kinematics

What students need to learn:

Kinematics of a particle moving in a straight line when the acceleration is a function of the displacement (x), or time (t).	The setting up and solution of equations where $\dfrac{dv}{dt} = f(t)$, $v\dfrac{dv}{dx} = f(x)$, $\dfrac{dx}{dt} = f(x)$ or $\dfrac{dx}{dt} = f(t)$ will be consistent with the level of calculus required in units C1, C2 and C4.

2 Elastic strings and springs

What students need to learn:

Elastic strings and springs. Hooke's law.	
Energy stored in an elastic string or spring.	Simple problems using the work-energy principle involving kinetic energy, potential energy and elastic energy.

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3 Further dynamics

What students need to learn:

Newton's laws of motion, for a particle moving in one dimension, when the applied force is variable.	The solution of the resulting equations will be consistent with the level of calculus in units C2, C3 and C4. Problems may involve the law of gravitation, ie the inverse square law.
Simple harmonic motion.	Proof that a particle moves with simple harmonic motion in a given situation may be required (ie showing that $\ddot{x} = -\omega^2 x$). Geometric or calculus methods of solution will be acceptable. Students will be expected to be familiar with standard formulae, which may be quoted without proof.
Oscillations of a particle attached to the end of an elastic string or spring.	Oscillations will be in the direction of the string or spring only.

4 Motion in a circle

What students need to learn:

Angular speed.	
Radial acceleration in circular motion. The forms $r\omega^2$ and $\frac{v^2}{r}$ are required.	
Uniform motion of a particle moving in a horizontal circle.	Problems involving the 'conical pendulum', an elastic string, motion on a banked surface, as well as other contexts, may be set.
Motion of a particle in a vertical circle.	

5 Statics of rigid bodies

What students need to learn:

Centre of mass of uniform rigid bodies and simple composite bodies.	The use of integration and/or symmetry to determine the centre of mass of a uniform body will be required.
Simple cases of equilibrium of rigid bodies.	To include (i) suspension of a body from a fixed point, (ii) a rigid body placed on a horizontal or inclined plane.